

Phase 4 — Tenant Rebilling Spec

PREPARED BY Energy Hippo implementation team

Exercise **every** allocation and reconciliation path. Vehicle (owner decision): **both** a new mixed-use **Northlake Town Center** (commercial allocation) and **sub-metered Cedar Row** (residential allocation). Lights up all 14 Tenant Rebilling reports + rp_TenantMap.

Entry path (see [entry-path-and-gaps.pdf](#)): bill allocation config (web), WUI master-point config (web), and bill-generation config (WPF TenantRebillView) are **UI-DIRECT**; meter allocation has an on-demand **run** button. **Note:** full tenant-**bill generation** runs via a scheduled SQL Agent job (Job_BillAllocation), not an on-demand button (**UI-THEN-RUN**, engine must be running). **Gap:** tenant **accruals** have no UI (A2). The 7 allocation methods are selectable; the engine branches only for method 1 (%/amt) and 6 (aggregation) — others are method-1 rows with precomputed shares.

EEMSuite facts (verified): two modes — **Bill Allocation** (split a parent bill's cost to tenants) and **Bill Generation** (rate × meter use → tenant bill) — plus the lower-level **WUI point allocation** (split a master point's use to sub-points). **7 bill-allocation methods, 3 point-allocation methods** (Percentage-Based is catalog-only/unimplemented), 4 bill-generation types (Off/Manual/Auto/Auto-Regen), reconciliation via penny-true-up + rp_BillReconciliation (source = interval or bill data). Most allocation "methods" share one engine — the percentages are precomputed and stored as method-1 detail rows.

Source-model extension

Add: tenant_space (building, name, floor_area, occupancy), bill_allocation_config (master_meter, method 1–7, fiscal_window, combine_overhead), bill_allocation_detail (config, tenant_space, percent, fixed_amount, is_amount_fixed, overhead_pct/amt, seq), bill_allocation_map (master_bill, allocated_bill), bill_aggregation_map, pt_allocation_config (master_pt, alloc_pt, method 1–3, effective), pt_allocation_index (child, index_value, unit, effective), bill_gen_config (meter, scenario, gen_status 1–4, schedule_code, recon_source 1/2). New EEM company **Northlake Town Center** + site/buildings/tenants extend data/security/ + data/scenarios/ manifests.

A. Northlake Town Center (commercial)

New mixed-use site under a new EEM company northlake_town_center (companyType commercial_multi_tenant). Buildings + masters + tenants:

building	master meter	tenants (sub-meters / spaces)
Market Hall (retail)	tc_market_main electric	Anchor Store, 4 in-line shops, Food Court, Common Area
Tower (office)	tc_tower_main electric	Floors 1–4 (sub-metered), Landlord/house meter
Campus Annex (demand)	tc_annex_main demand-metered	Bldg A, B, C (usage + coincident demand)
Substation feed	tc_substation_master	Substation 1, Substation 2

building	master meter	tenants (sub-meters / spaces)
— (cost center)	tc_property_total	rolls up all tenant bills

Allocation method coverage (one config each → every method)

#	method	config
1	Fixed Amount & %	Market Hall → Anchor fixed \$2,000; Food Court 15%; in-line shops remaining % — with overhead in both combined (combine_overhead=1) and separate modes
2	Fully Submetered	Tower → 4 floors, each a sub-meter (consumption share)
3	Partially Submetered	Tower variant → 3 submetered floors + landlord/house meter for the remainder
4	Building Areas	Market Hall in-line shops → share by tenant_space.floor_area
5	Partially Submetered w/ Coincident Demand	Campus Annex → Bldg A/B/C by usage + coincident peak
6	Bill Aggregation	tc_property_total ← sum of all tenant bills (children → parent, bill_aggregation_map)
7	Metered Substations	Substation feed → Substation 1/2 by % of usage

B. Cedar Row sub-metering (residential)

Cedar Row A and Cedar Row B are master-metered: Valley Electric bills Cedar Row A Electric Master and Cedar Row B Electric Master, River City bills Cedar Row A Water Master and Cedar Row B Water Master, and no resident gets a utility bill. Every apartment already has an owned electric submeter and an owned water submeter behind those masters (Cedar Row A: 60 apartments, units 101–120, 201–220 and 301–320; Cedar Row B: 64 apartments, units 101–416), one row per apartment on the workbook tab **Units and Submeters**. Rebilling maps apartments to these submeter points; it adds no meters.

- **Electric: method 2 (fully submetered).** Each apartment is allocated its building's electric master bill by its own electric submeter: meters Cedar Row A Unit 101 Electric through Cedar Row B Unit 416 Electric (tags SYN-SUB-E-A101 ... SYN-SUB-E-B416), hourly point Submeter kWh 60m from the fixed-network electric submeters.
- **Water and sewer: method 2.** Each apartment is allocated water and sewer by its own water submeter (tags SYN-SUB-W-A101 ... SYN-SUB-W-B416), read on the monthly Neptune apartment water route with points Register Reading and Route Usage gal. The apartment's sewer share is capped at its winter average.
- **House submeters: overhead.** Cedar Row A Electric House, Cedar Row A Water House, Cedar Row B Electric House and Cedar Row B Water House cover corridors, laundry and lighting; their usage is recovered as **overhead**, not charged to one apartment.
- **Reconciliation.** For each master bill, the apartment charges plus overhead sum to the master bill. The aggregates Cedar Row A Submetered kWh Total and Cedar Row B Submetered kWh Total (every apartment and house electric submeter in the building) are compared with the electric master bill for unaccounted load.
- The residential case complements the commercial Town Center: fully submetered allocation with overhead on a housing community.

C. WUI point families (master point → sub-points)

Separate from bill allocation; needs interval + index + area + a generation rate.

family	method	setup
wui_market_electric	1 Weighted Use Index	master electric point + 3 child points each with pt_allocation_index (kWh/sqft-day) + area + bill_gen_config rate
wui_tower_electric	2 Measured Use + residual	master + one measured child (own interval, subtracted) + one WUI child (gets residual)

Deliberate fail-reason fixtures (PtAllocFailReason): a child with **no index** (reason 2); one with **no area** (reason 3); a master/child with **partial interval** (reason 1); a 3-level chain where the mid master fails so the leaf inherits **reason 4**; a valid child with **no rate** (reason 5). Plus a **method-3 (Percentage Based)** child to demonstrate it is **silently ignored** (documented product gap).

D. Bill generation

Assign a multi-component rate (Phase 2 determinants + UtilityCommodityMap to ≥2 charge commodities) to a tenant meter under both an **Actual (-1)** and a **non-actual** scenario. Replicate bill_gen_config four times with gen_status = **1 Off / 2 Manual / 3 Auto / 4 Auto-Regen**; schedule code **CC** (Company Calendar). Seed ≥3 periods of interval data so the generator produces UtilityBillGenLog + child bills. Lights up GeneratedBills, BillGenerationConfig, BillMaintenance, RateComponentMap.

E. Reconciliation

Give one Town Center meter a **Bill Meter Reconciliation** scenario (ScenarioTypeId 20): recon_source=1 (interval) on some meters, =2 (bill) on others. Build a **deliberate rate/usage mismatch** on one meter so rp_BillReconciliation shows nonzero amount/quantity variance (over/under recovery). The allocation path's penny-true-up still makes child bills sum to the parent.

F. Tenant accruals

A tenant account missing its current-period bill, with a manual 2300-tagged bill (open accrual) + a trued-up accrual↔reversal pair — feeding rp_accruals / rp_mpaaccruals in the tenant context.

expectedAssertions

- every_alloc_method_configured — one bill_allocation_config per method 1–7 exists and produces tenant bills.
- children_sum_to_parent — for each allocation, $\sum$ child bill amounts = parent bill amount (penny-true-up), incl. overhead lines.
- aggregation_rolls_up — tc_property_total bill = $\sum$ consumed child bills (bill_aggregation_map).
- wui_split_sums_to_master — WUI family child allocated use = master residual, proportional to index × area × days.

- `wui_fail_reasons` — each of fail reasons 1–5 is produced by its fixture; the method-3 child yields no output and no failure.
- `bill_gen_types` — Off/Manual/Auto/Auto-Regen behave per type; Auto creates missing-period bills, Auto-Regen regenerates on new data.
- `reconciliation_variance` — `rp_BillReconciliation` shows the deliberate nonzero variance with interval vs bill source.
- `tenant_accrual_lifecycle` — open tenant accrual listed, then trued up to zero.